Syllabus Year: 2024-2025

**CHRIST (Deemed to be University), Bengaluru - 560029**

**END-SEM EXAMINATIONS - APRIL 2025**

**Bachelor of Computer Applications SEMESTER - II**

**Course Code: BCA101-2** **Max Marks: 100**

**Course Name: DISCRETE MATHEMATICS** **Duration: 03:00**

**General Instructions:**

- All rough work should be done in the answer script. Do not write or scribble in the question paper except your register number.
- Verify the Course code / course title and number of pages of questions in the question paper.
- Make sure your mobile phone is switched off and placed at the designated place.
- Malpractices will be viewed very seriously.
- Answers should be written on both sides of the paper in the answer book. No sheets should be detached from the answer book.
- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer book.

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**Course Outcome:** Students will be able to:

CO1: Demonstrate a working knowledge of set notation and elementary set theory.

CO2: Apply the different properties of injections, surjections, bijections, compositions, and inverse functions.

CO3: Understand different types of matrices and operations among matrices.

CO4: Determine when a relation is reflexive, symmetric, antisymmetric, or transitive, apply the properties of equivalence relations and partial orderings, and explain the connection between equivalence relations.

CO5: Effectively express the elementary concepts and results of Number theory

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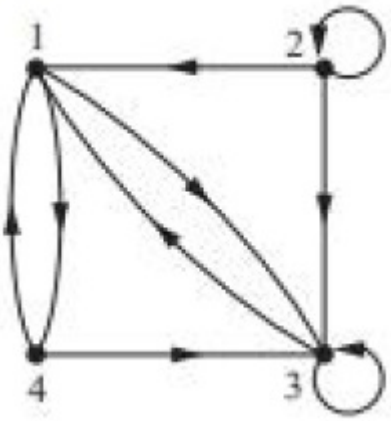
**Section A** **3*8=24**

Answer ALL the questions

| | | | |
|---|---|---|---|
| **1)** | What is the Cardinality of each of these sets?<br>(a) $\{\phi\}$<br>(b) {{a}, {b}, {c}}<br>(c) {{b}, {b, {b}}} | **3** | (CO1, L2) |
| **2) a)** | Find the values of $(a)\ \lceil \lfloor \frac{1}{9} \rfloor + \lceil \frac{1}{3} \rceil + \frac{1}{6} \rceil \quad (b)\ \lceil \frac{1}{8} + \lfloor \frac{1}{3} \rfloor \rceil \quad (c)\ \lfloor \frac{1}{8} + \lceil \frac{7}{3} \rceil \rfloor$. | **3** | (CO2, L3) |
| | **OR** | | |
| **b)** | Determine whether $f$ is a function from $Z$ to $R$ if $(a)\ f(x) = \pm x, \quad (b)\ f(x) = \sqrt{x^2+1}, \quad (c)\ f(x) = \frac{1}{x^2-4}$. | **3** | (CO2, L3) |
| **3)** | Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 3 & 6 \\ -1 & 2 \end{pmatrix}$, find $A^2 + 7B$. | **3** | (CO3, L3) |
| **4)** | Give an example of a relation on a set is<br>(a) both symmetric and antisymmetric.<br>(b) neither symmetric nor antisymmetric. | **3** | (CO4, L3) |
| **5) a)** | Obtain the relation and then draw the directed graph for the relation matrix $M_R = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$. | **3** | (CO4, L3) |
| | **OR** | | |

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END-SEM EXAMINATIONS - APRIL 2025

**b)** What are the ordered pairs in the relation R represented by the directed graph and also represent the same relation in the matrix form? **3** (CO4, L3)




**6)** Solve the congruence $5x \equiv 7 (mod\, 19)$. **3** (CO5, L2)

**7)** Prove or disprove: For integers, $a,\ b,$ and $c,$ if $a|b,$ and $b|c,$ then $a|c.$ **3** (CO5, L2)

**8) a)** Explain congruence modulo $m$. Determine whether 17 is congruent to 5 modulo 6 and whether 24 and 14 are congruent modulo 6. **3** (CO5, L3)

**OR**

**b)** What is the decimal expansion of $(2AE0B)_{16}$? **3** (CO5, L3)

**Section B** **8*7=56**

Answer ALL the questions

**9)** Use set-builder notation and logical equivalencies to establish the De Morgan's laws. **8** (CO2, L2)

**10) a)** (a) Prove that if $x$ is a real number then $\lfloor 2x \rfloor = \lfloor x \rfloor + \lfloor x + \frac{1}{2} \rfloor$. **8** (CO3, L4)

(b) Determine whether the function $f(x) = x^4$ is bijection from R to R?

**OR**

**b)** Define invertible function. Let $f : R \to R$ is defined by $f(x) = ax + b$ for $a, b \in R$ and $a \neq 0$. Show that $f$ is invertible and find the inverse of $f$. Suppose if $a = 0$ then the function is invertible or not. Justify your answer. **8** (CO3, L4)

**11)** Let $A = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{pmatrix}$. Find the $n^{th}$ Boolean power of A that is $A^{[n]}$ for all positive integers $n$. **8** (CO3, L4)

**12)** Which of these relations on {0, 1, 2, 3} are equivalence relations? Determine the properties of an equivalence relation that the others lack. **8** (CO4, L4)

$R_1 = \{ (0,0), (1,1), (2,2), (3,3) \}$
$R_2 = \{ (0,0), (0,2), (2,0), (2,2), (2,3), (3,2), (3,3) \}$
$R_3 = \{ (0,0), (1,1), (1,2), (2,1), (2,2), (3,3) \}$
$R_4 = \{ (0,0), (1,1), (1,3), (2,2), (2,3), (3,1), (3,2), (3,3) \}$
$R_5 = \{ (0,0), (0,1), (0,2), (1,0), (1,1), (1,2), (2,0), (2,2), (3,3) \}$

**13) a)** (a) Show that the inclusion relation $\subseteq$ is a partial ordering on the power set of a set S i.e., P(S). **8** (CO4, L3)

(b) Let $R$ be the relation on set of people such that $xRy$ if $x$ and $y$ are people and $x$ is older than $y$. Show that $R$ is not a partial ordering.

**OR**

**b)** Define equivalence relation with example. Let R be a relation on the set of real numbers such that $xRy$ if and only if $x$ and $y$ are real numbers that differ by less **8** (CO4, L3)

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END SEM EXAMINATIONS - APRIL 2025

than 1, that is $|x - y| < 1$. Verify whether R is equivalence relation or not.

**14)** What is the greatest common divisor of 123 and 2347? Express it as a linear combination of 123 and 2347. **8** (CO5, L2)

**15) a)** Write the Euclidean algorithm and determine the greatest common divisor of 414 and 662. Also find their least common multiple. **8** (CO5, L4)

**OR**

**b)** Describe the algorithm on modular exponentiation and find $123^{1001}\ mod\ 101$. **8** (CO5, L4)

**Section C** **10*2=20**

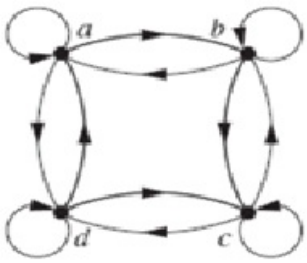
Answer ALL the questions

**16) a)** Let $R_1$ and $R_2$ are relations where $R_1$ is a relation from {1, 2, 3} to {1, 2, 3, 4} with $R_1 = \{(1,1), (1,4), (2,3), (3,1), (3,4)\}$ and $R_2$ is the relation from {1, 2, 3, 4} to {0, 1, 2} with $R_2 = \{(1,0), (2,0), (3,1), (3,2), (4,1)\}$. Find $R_1 \cup R_2,\ R_1 \cap R_2,\ R_1 - R_2,\ R_2 - R_1,$ and $R_1 \oplus R_2$ and its corresponding relational matrix. **10** (CO4, L3)

**OR**

**b)** (a) Draw the Hasse diagram for the poset $(S, \subseteq)$ where $S = \{a, b, c\}$and write its greatest and least elements if it exists. **10** (CO4, L3)

(b) Determine whether the relation represented by the following digraph is partial ordered relation. Justify your answer.




**17)** (a) Find all the solutions to the system of congruences $x \equiv 2(mod 3),\ x \equiv 1(mod 4)$ and $x \equiv 3(mod 5)$. **10** (CO5, L2)

(b) What is the message produced from the message MEET YOU IN THE PARK using the Caesar cipher.

| **Revised Bloom's Taxonomy (RBT) Levels** | | |
|---|---|---|
| L1 - Remember | L2 - Understand | L3 - Apply |
| L4 - Analyse | L5 - Evaluate | L6 - Create |

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